

Assignment 1: Recap of Differential Geometry and General Relativity

to be handed in online by Sunday, April 26th

Students taking the “Classical and Quantum Aspects of Black Holes” physics course are only expected to submit the exercises marked with an asterisk*, but are obviously invited to try all of them. Students taking the “Analysis and Geometry of Classical Systems” maths course are expected to submit all exercises.

If you decide to skip an exercise, be sure to read it at least, since you will possibly need its result for some later exercise.

We have hidden some hints as white text to encourage you to try to solve the problems without looking at them. If you are stuck, simply copy the white text somewhere else to reveal it.

Exercise 0: Recap of Diff Geo

Recall that for vector fields $X, Y, Z \in \mathfrak{X}(\mathcal{M})$ and smooth functions $f, g \in \mathcal{C}^\infty(\mathcal{M})$, the connection ∇ satisfies:

1. $\nabla_{fX+gY}Z = f\nabla_XZ + g\nabla_YZ$,
2. $\nabla_X(Y+Z) = \nabla_XY + \nabla_XZ$,
3. $\nabla_X(fY) = (\nabla_Xf)Y + f\nabla_XY$, with $\nabla_Xf = X(f)$.

Let x^μ be a local coordinate system, with associated coordinate basis $\{\partial_\mu\}$. We define the Christoffel symbols as $\nabla_{\partial_\mu}\partial_\nu = \Gamma_{\mu\nu}^\rho\partial_\rho$. Now let $V = V^\mu\partial_\mu$ and $W = W^\nu\partial_\nu$ be two arbitrary vector fields, and let $\omega = \omega_\mu dx^\mu$ be a covector field.

a) Using the axioms and definitions above, prove

$$\nabla_V W = V^\mu (\partial_\mu W^\rho + \Gamma_{\mu\nu}^\rho W^\nu) \partial_\rho. \quad (1)$$

Then, recalling that $\omega(V) = \omega_\mu V^\mu$, prove that

$$\nabla_V \omega = V^\mu (\partial_\mu \omega_\nu - \Gamma_{\mu\nu}^\rho \omega_\rho) dx^\nu. \quad (2)$$

Identify the expressions in brackets.

b) Using the identity $\Gamma_{\mu\nu}^\mu = \partial_\nu \ln \sqrt{|g|}$, where $g = \det(g_{\mu\nu})$, show that

$$\nabla_\mu V^\mu = \frac{1}{\sqrt{|g|}} \partial_\mu (\sqrt{|g|} V^\mu). \quad (3)$$

Exercise 1: Causal Vectors

A tangent vector V^μ is *causal* if it is timelike or lightlike, that is, if $V^\mu V_\mu = g_{\mu\nu} V^\mu V^\nu \leq 0$. Then, we say that a spacetime $\mathcal{M}$ is *time-orientable* if it admits a *time-orientation*: an everywhere causal vector field T^μ . At any $p \in \mathcal{M}$, the metric determines two light-cones in $T_p\mathcal{M}$. We can then define the light-cone in which T^μ lies as the “future” light-cone, and the other one as the “past” light-cone. Then, we say that any causal vector X^μ is *future-directed* if it lies in the same light-cone as T^μ , and *past-directed* otherwise.

a)* Let X^μ and Y^μ be two causal vectors at a point $p \in \mathcal{M}$ such that X^μ and Y^μ are not proportional to each other. Show that

$$X_\mu Y^\mu \leq 0 \iff X^\mu \text{ and } Y^\mu \text{ lie in the same light-cone.} \quad (4)$$

Hint:

Exercise 2: Scalar Fields and the Stress-Energy Tensor

The action for a minimally-coupled scalar field ϕ is

$$S_M = - \int_{\mathcal{M}} d^4x \sqrt{-g} \left[\frac{1}{2} \partial_\mu \phi \partial^\mu \phi + V(\phi) \right], \quad (5)$$

where $V(\phi)$ is the potential of the scalar field.

a)* Imposing Dirichlet boundary conditions, $\delta\phi|_{\partial\mathcal{M}} = 0$, vary S_M with respect to ϕ to find the Klein-Gordon equation,

$$\square\phi - V'(\phi) = 0, \quad (6)$$

where $\square := \nabla_\mu \nabla^\mu$. Do not use the Euler-Lagrange equations, but rather show explicitly how to deal with the boundary term.

b)* Vary S_M with respect to the inverse metric $g^{\mu\nu}$ to find the stress-energy tensor,

$$T_{\mu\nu} = \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} \nabla^\rho \phi \nabla_\rho \phi - g_{\mu\nu} V(\phi). \quad (7)$$

Exercise 3: Cosmology I¹

The flat FLRW metric, describing a homogeneous and isotropic flat Universe, reads

$$ds^2 = -dt^2 + a^2(t) (dx^2 + dy^2 + dz^2). \quad (8)$$

Notice that the x, y, z coordinates do not measure *physical* distances. They are called *comoving coordinates*, since they “move” with the expansion of the Universe.

a)* Derive the geodesic equation for particles in a FLRW universe after arguing that we can focus solely on motion along the x direction without loss of generality. Argue that the energy E of massless particles decays with the expansion of the Universe. Comment on the physical relevance of this result.

Hint: Recall that E is the 0th component of the 4-momentum, so $E \propto dt/d\sigma$ with σ being an affine parameter along the geodesic.

b) Read off the Christoffel symbols from the geodesic equation. Then compute the Ricci tensor and the Ricci scalar. You may check your results using `diffgeo`.

Hint: Diagonal metrics that depend only on one coordinate have a diagonal Ricci tensor.²

¹To learn more about this fascinating topic, we recommend the excellent lecture notes by Daniel Baumann.

²You can find a proof of this fact in [gr-qc/9602015].

Exercise 4: Energy Conditions

a) Prove that DEC $\implies$ WEC.

Hint:

b)* The stress-energy tensor of a perfect fluid reads

$$T_{\mu\nu} = (\rho + p)U_\mu U_\nu + pg_{\mu\nu}, \quad (9)$$

where U^μ , a future-directed timelike vector with $U^\mu U_\mu = -1$, is the fluid's 4-velocity, and ρ and p are the fluid's energy density and isotropic pressure, respectively.

Show that

$$\text{NEC} \implies (\rho + p) \geq 0, \quad (10)$$

$$\text{WEC} \implies \rho \geq 0 \quad \text{and} \quad (\rho + p) \geq 0, \quad (11)$$

$$\text{DEC} \implies \rho \geq |p|, \quad (12)$$

$$\text{SEC} \implies (\rho + 3p) \geq 0 \quad \text{and} \quad (\rho + p) \geq 0. \quad (13)$$

Hint: